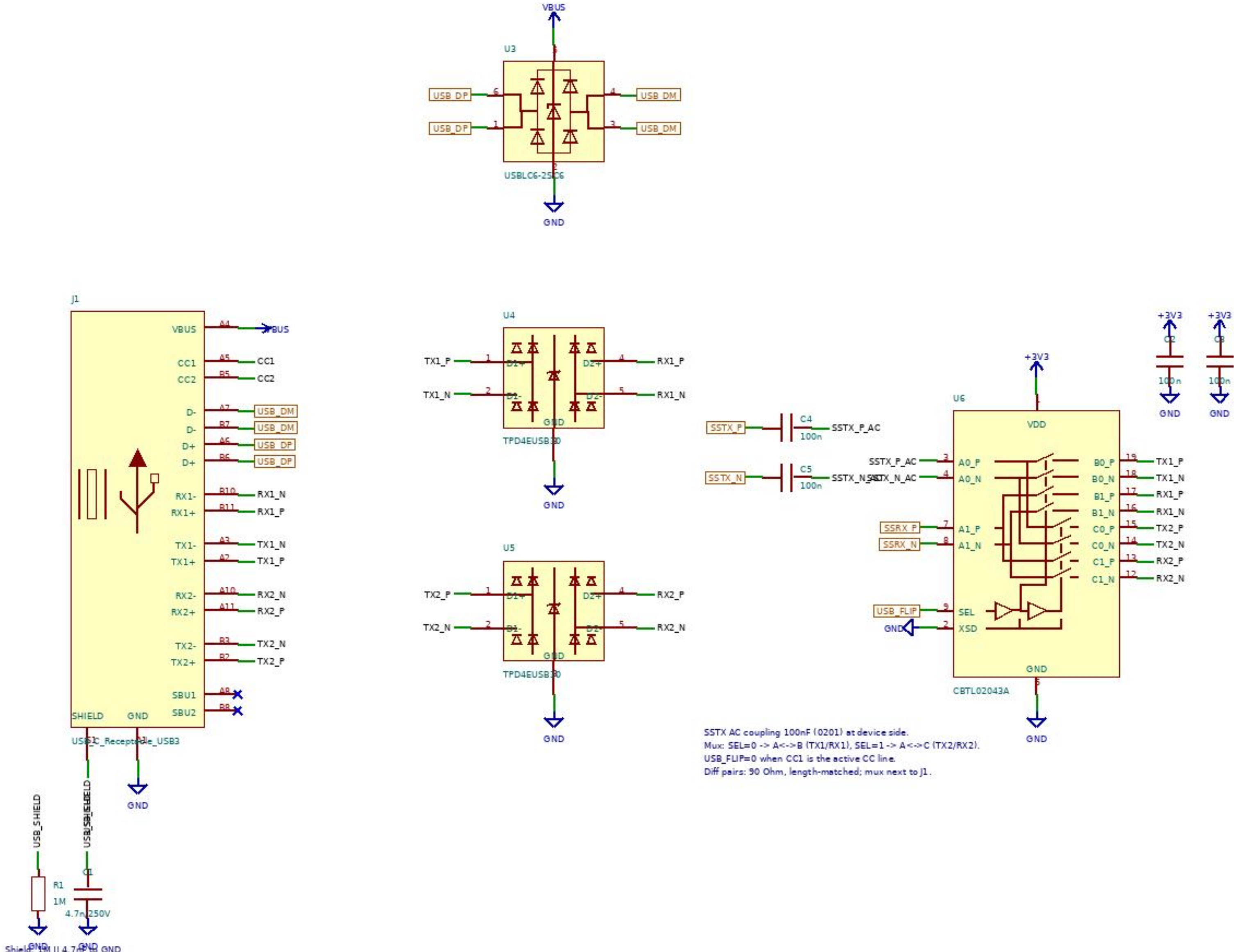


Open Debug + Trace probe, rev 0.1 (direction B)

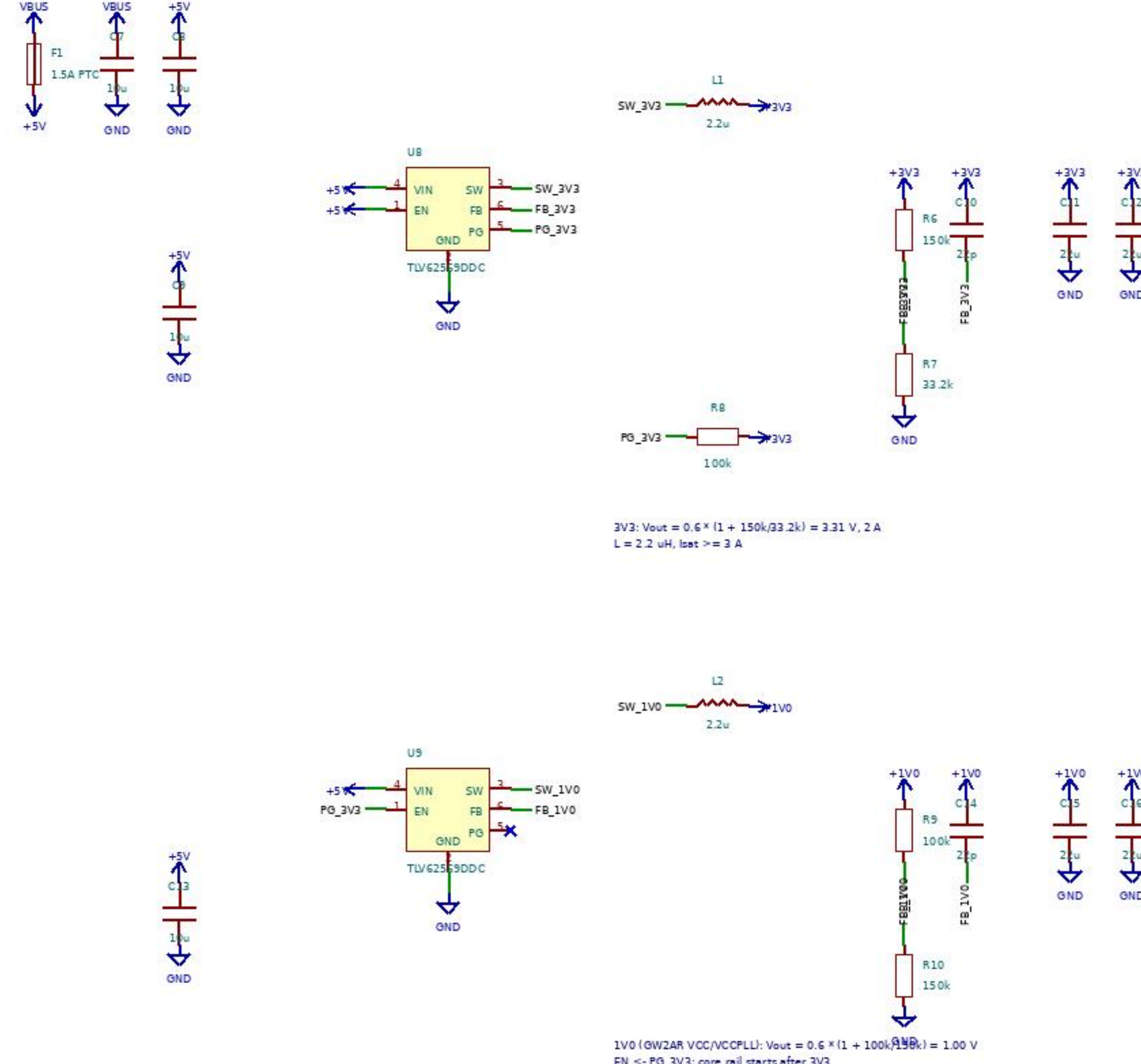
GW2AR-18 QN88 (trace capture, timestamps, SWD/JTAG engine, 64 Mbit SDRAM buffer)
+ CH569W (USB3 SS device; ORBTrace-compatible trace IF, events IF, CM9IS-DAP v2, CDC, DFU).
Target: MIP120, VREF 1.2..3.6 V. Triggers: 2x SMA. Spec: see design document.

Sources of pinouts: GW2AR-18 - Oswin UQ115E v1.7.2 (QN88, SDRAM); CH569W - WCH reference schematic CH569W-R0-1v0; other ICs - KiCad library symbols. Footprints marked probe: are project-local.
Open items before layout: CH569 EP size (datasheet), QN88 land pattern (UQ229 drawing), CBTL02043A SEL polarity, USB-C part, MODE straps (UQ290), PB15 RST# config.

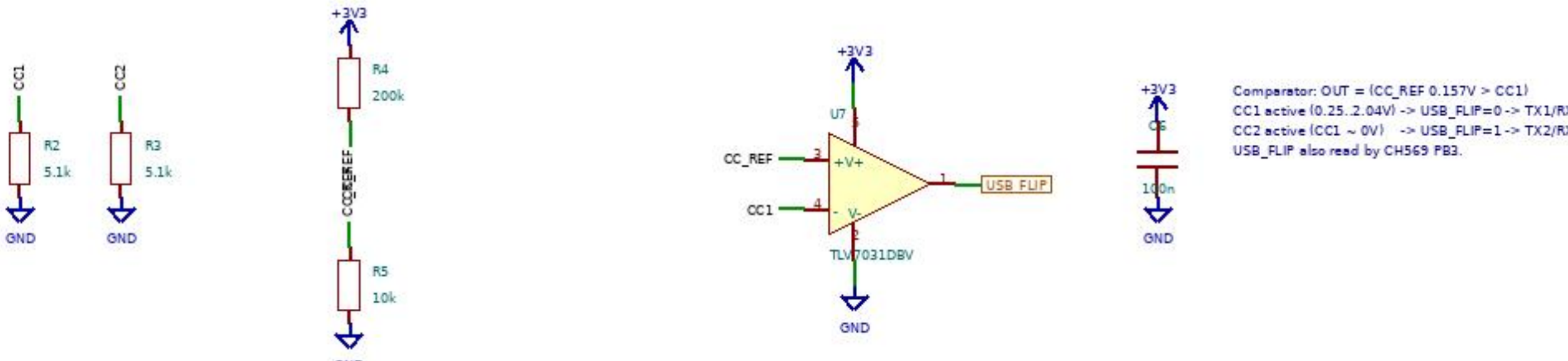
USB-C 3.2 Gen1 receptacle, SS orientation mux, ESD



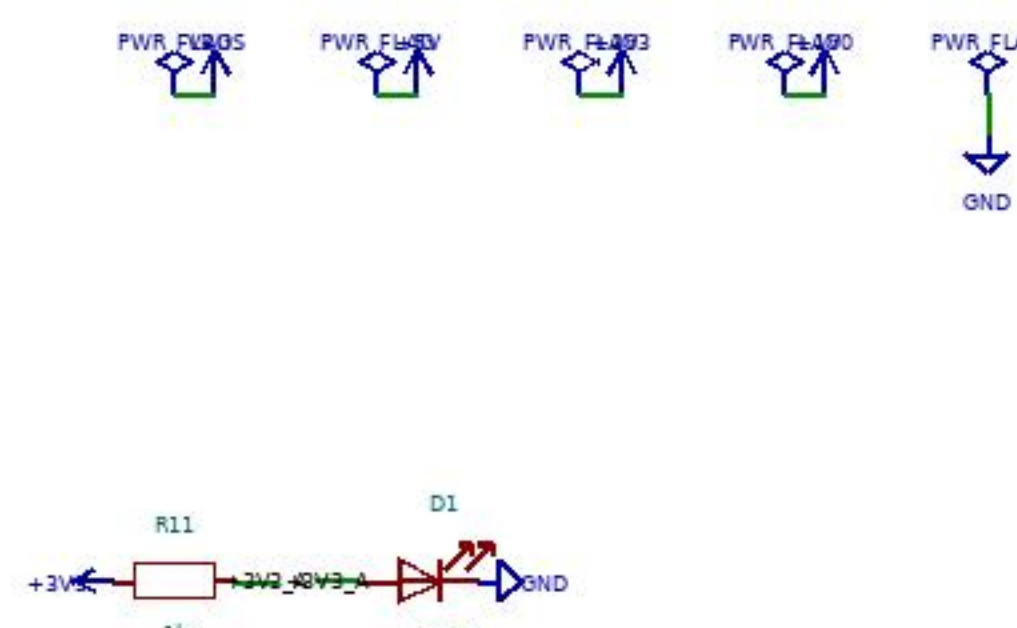
Power: VBUS -> 5V -> 3V3 / 1V0 (TLV62569)



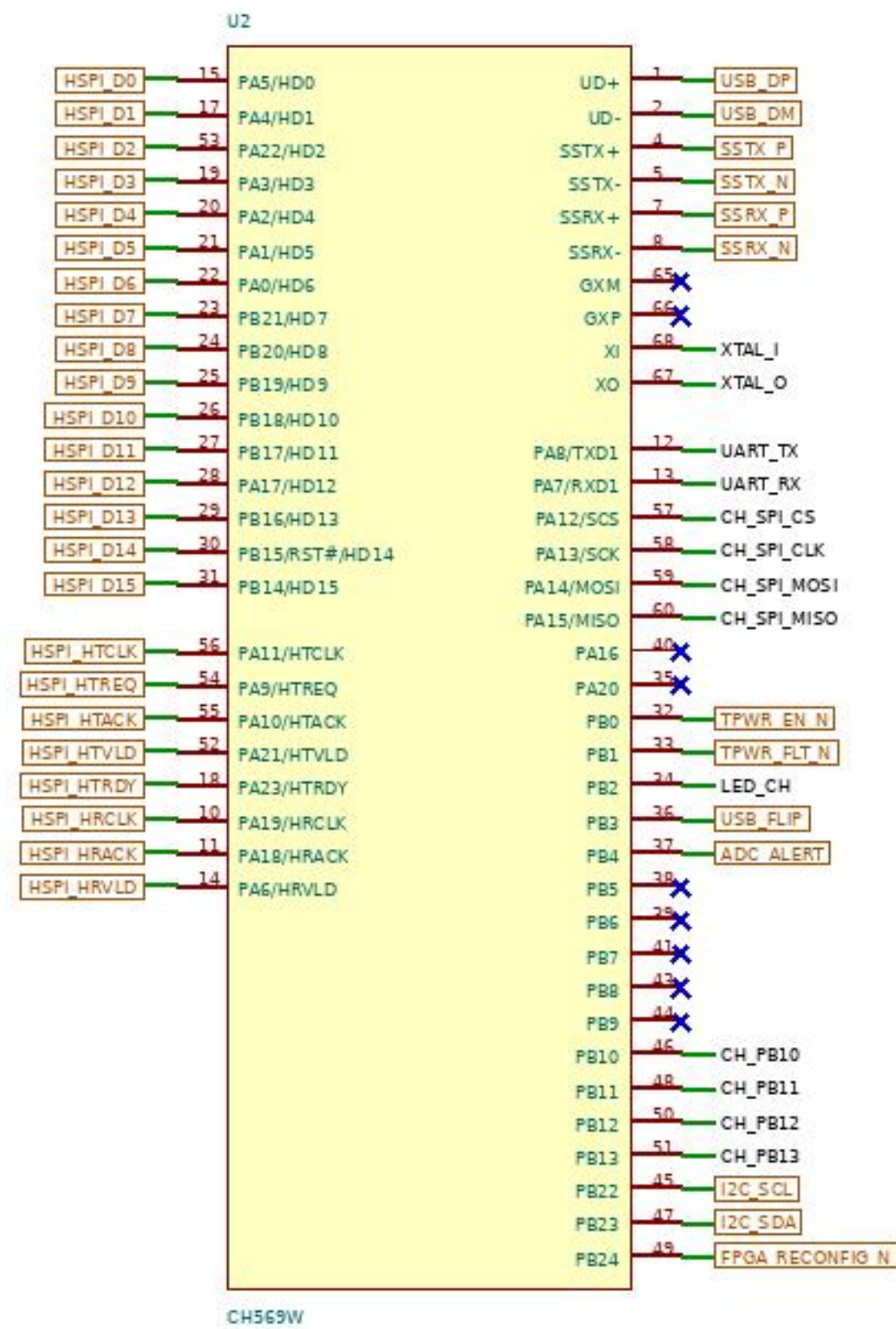
CC orientation detect (UFP, Rd = 5.1k)



Power flags, indication

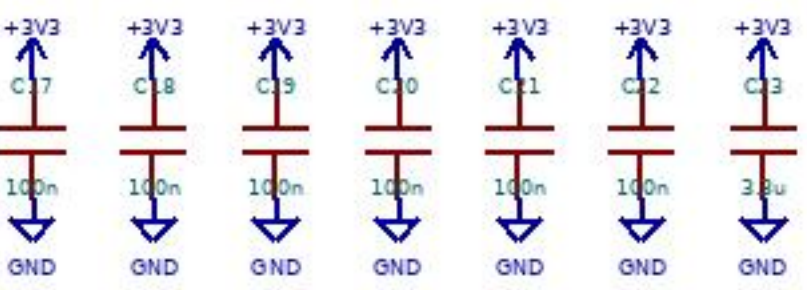
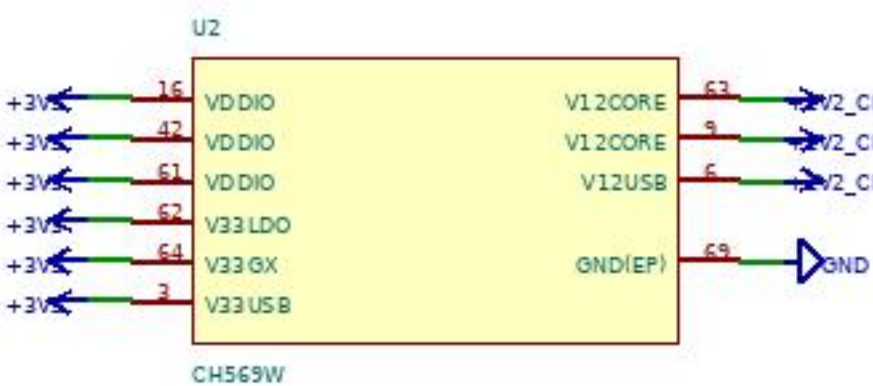


CH569W: USB3 SS device, HSPI 16 bit to FPGA

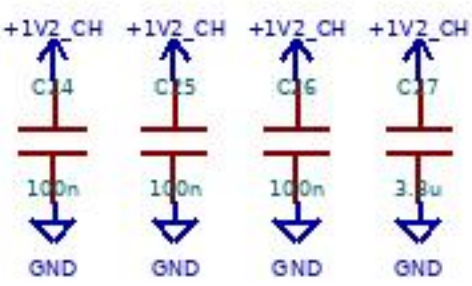


PB15 = HD14 is also RST#: disable the RST# function in the CH569 config (WCH ISP tool) before the FPGA drives HSPi_D14. 10k pull-up keeps it inactive. HD0 low at power-up = USB ISP bootloader (BOOT button).

Power decoupling (per WCH CH569W-R0 reference)

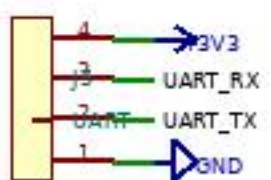
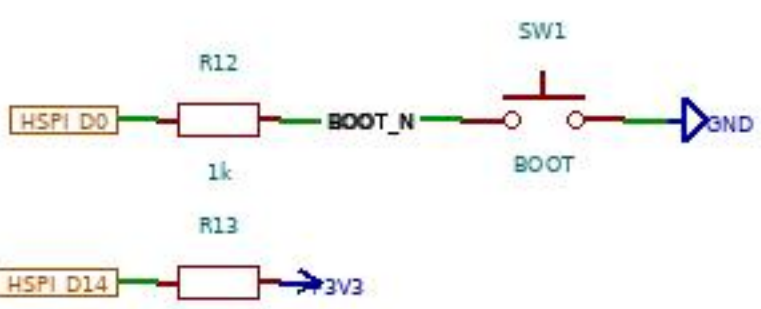
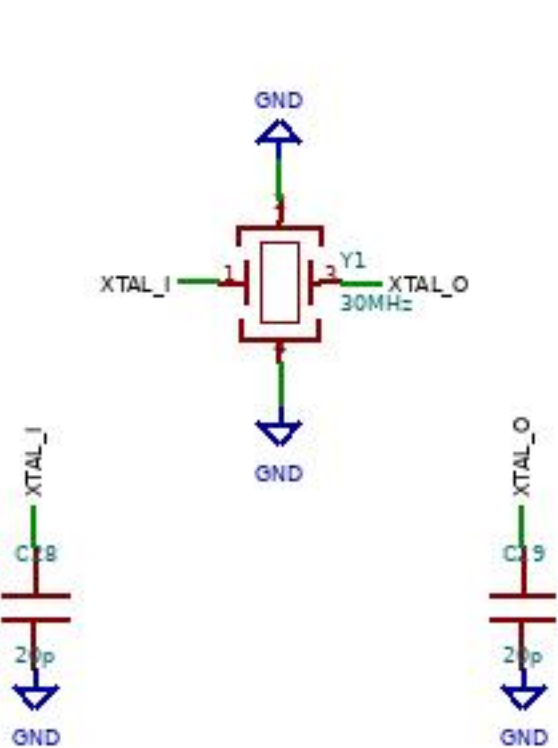


VDDIO x3, V33LDO, V33GX, V33US8: 100n each + 3.3u at V33LDO

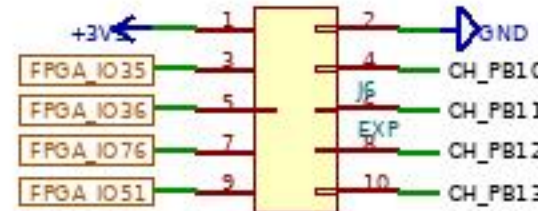
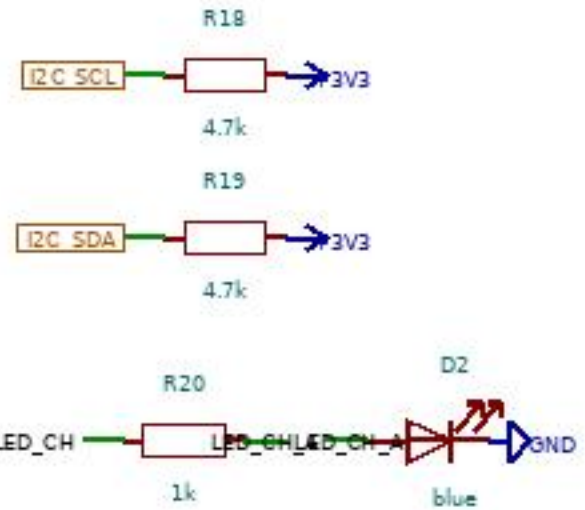
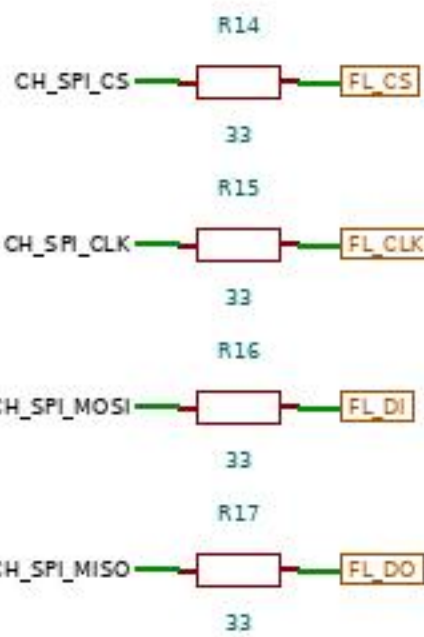


+1V2_CH = internal LDO output (V12CORE pin 63): 100n at pins 9, 63, 6 + 3.3u at pin 63. Do not load externally.

Clock, boot, UART, FPGA-flash access, I2C, expansion



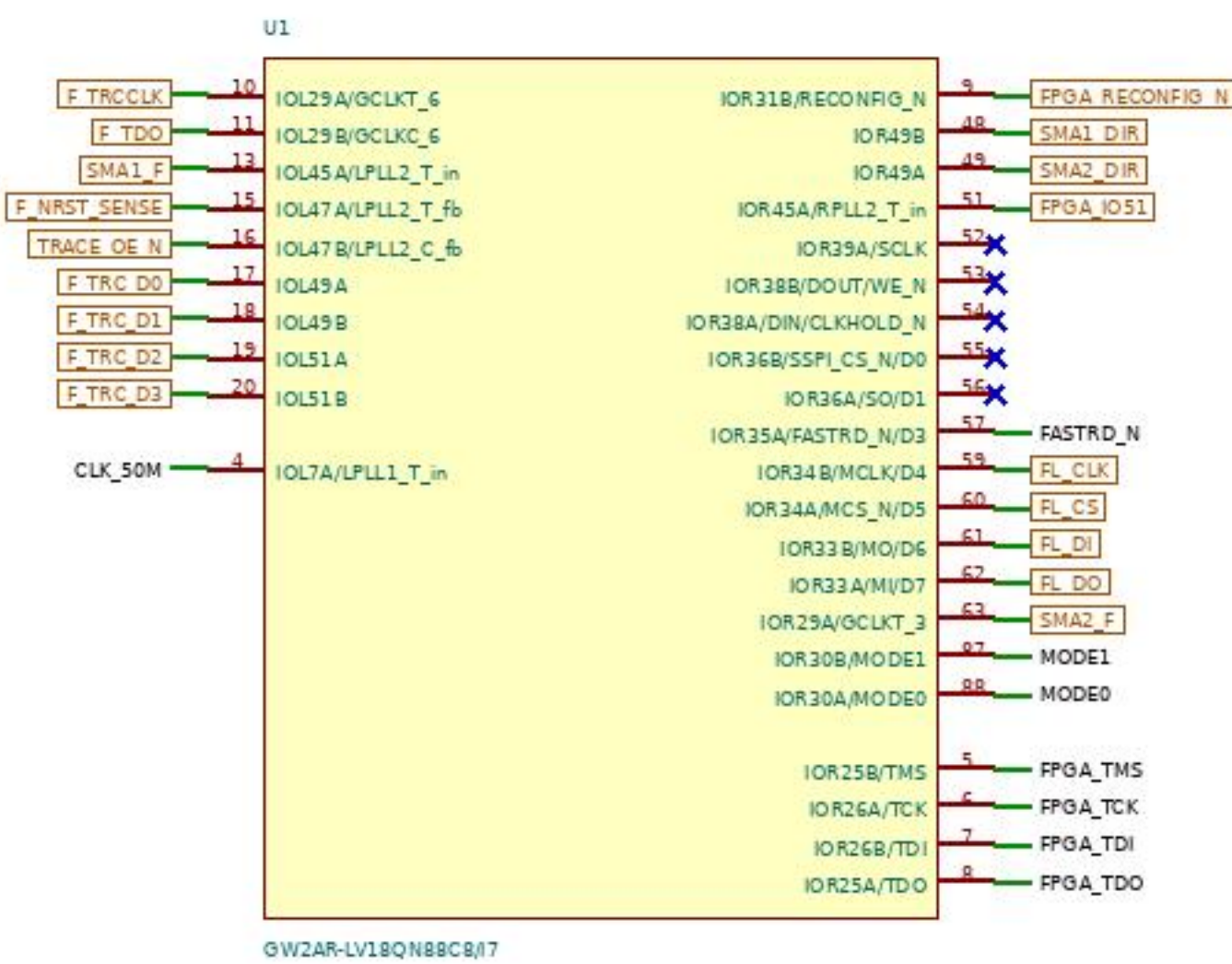
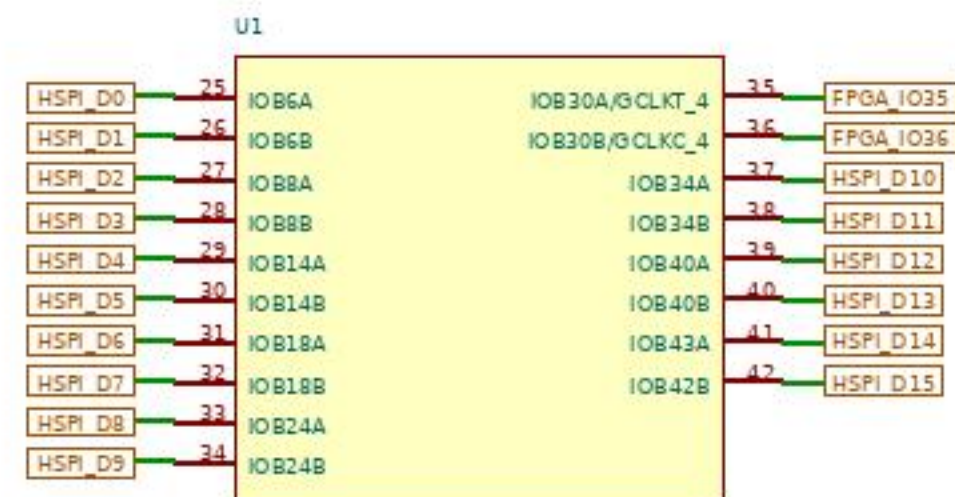
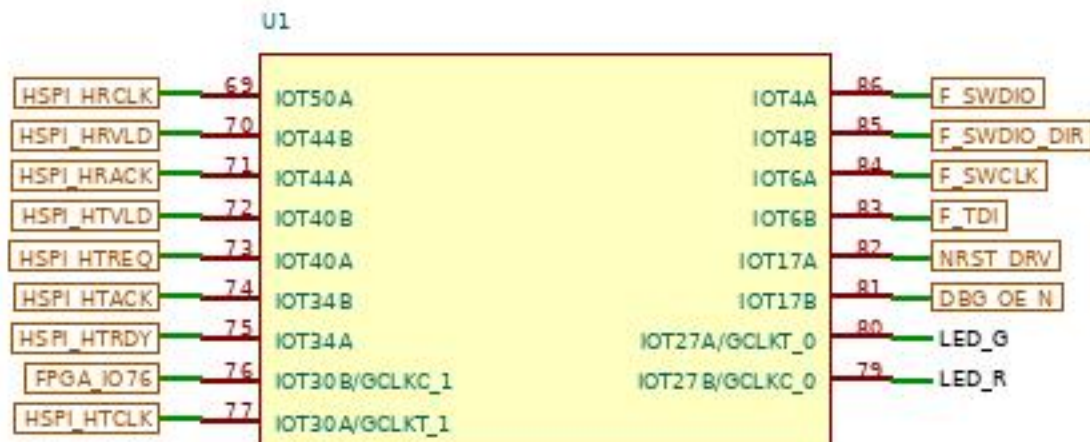
UART1: 1 GND, 2 TX (out), 3 RX (in), 4 3V3



EXP: spare FPGA + CH569 GPIO (debug, test)

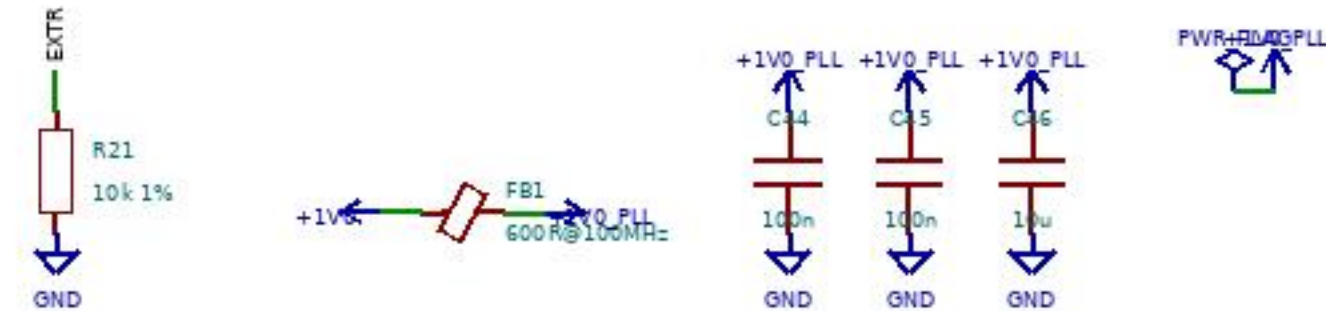
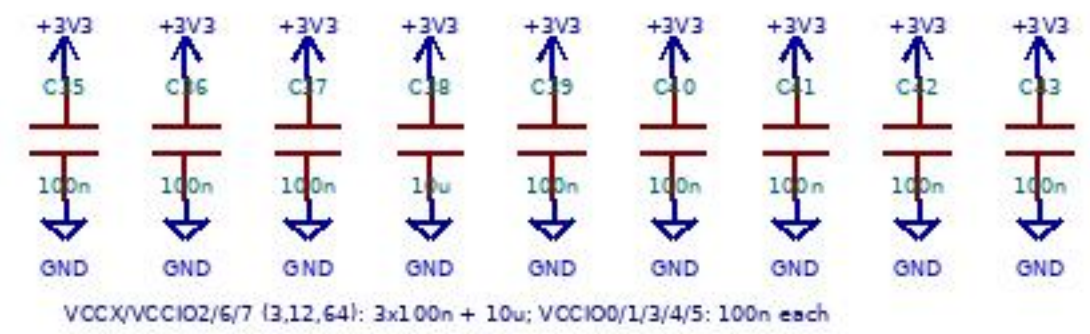
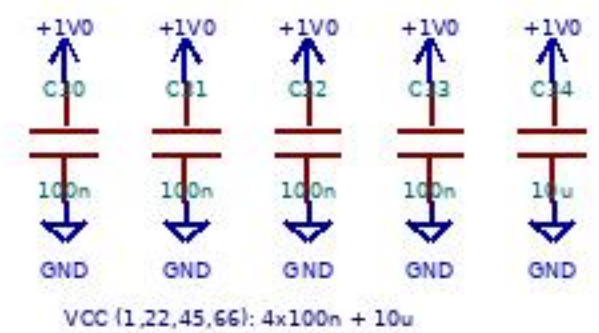
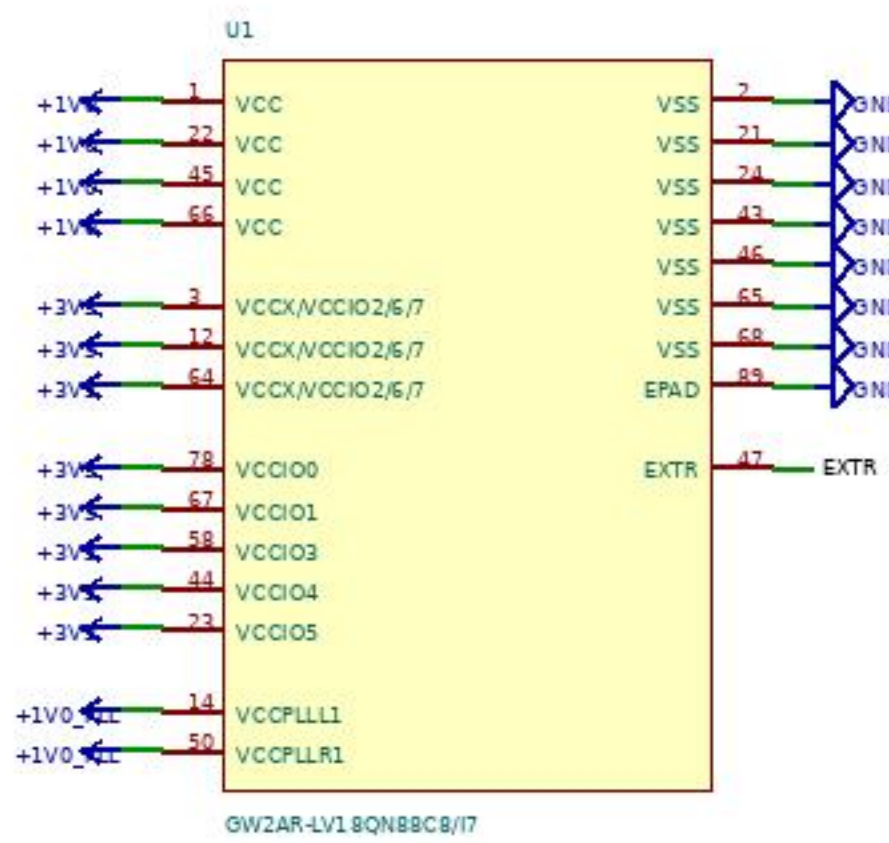
CH569 SPI0 -> FPGA config flash (update path). CH569 drives only while FPGA_RECONFIG_N = 0; otherwise SPI0 pins stay inputs (Hi-Z).

GW2AR-LV18QN88C8/I7 I/O banks (all VCCIO = 3.3 V)



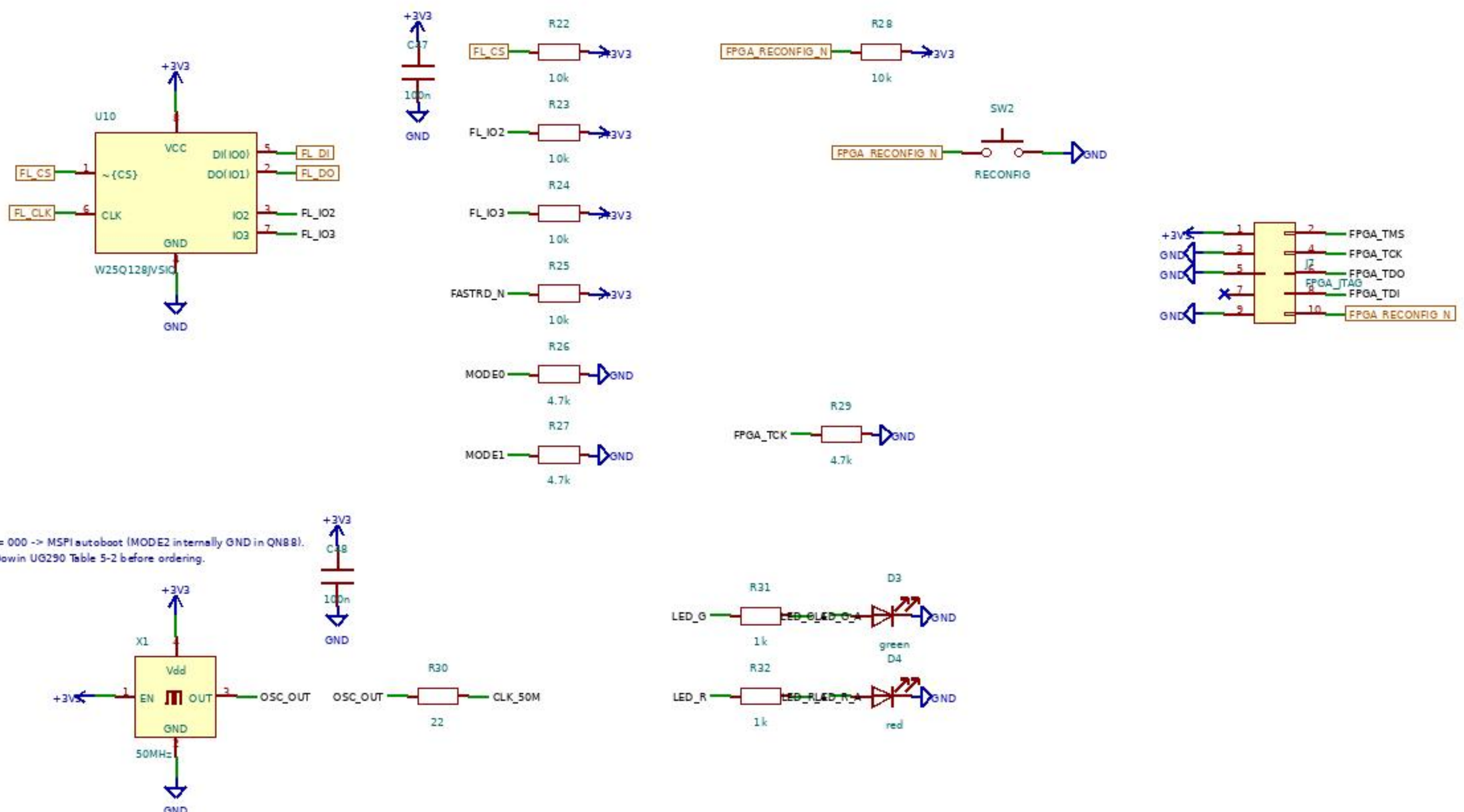
Trace port on bank 6: TRC_CLK on GCLKT_6 (pin 10), data on 17..20 (IDDR + IODELAY).
Timestamp ref: SMA1 on LPLL2_T_in (13), local 50 MHz on LPLL1_T_in (14).
HSP1: data on banks 5/4, CH569 HTCLK on GCLKT_1 (77).

Power and decoupling (UG115: pins 3/12/64 = VCCX/VCCIO2/6/7)



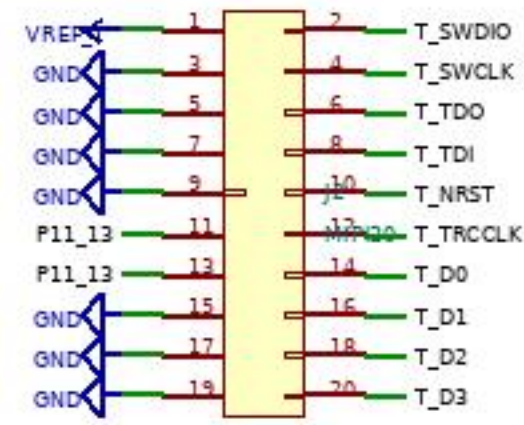
VCCPLL1 (14), VCCPLLR1 (50): 1.0 V via ferrite. EXTR (47): 10k 1% to GND (UG115).

Configuration (MSPi flash), JTAG, clock, LEDs

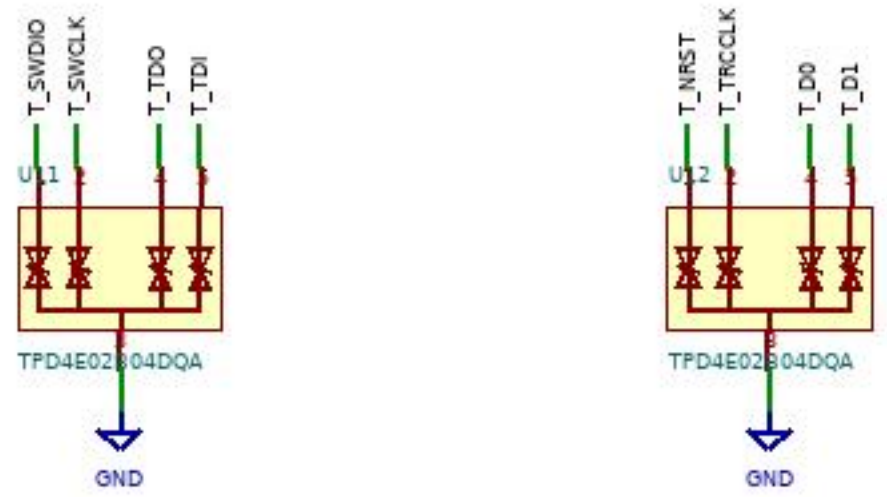


MODE[2:0] = 000 -> MSPI autoboot (MODE2 internally GND in QN88).
Verify with Gowin UG290 Table 5-2 before ordering.

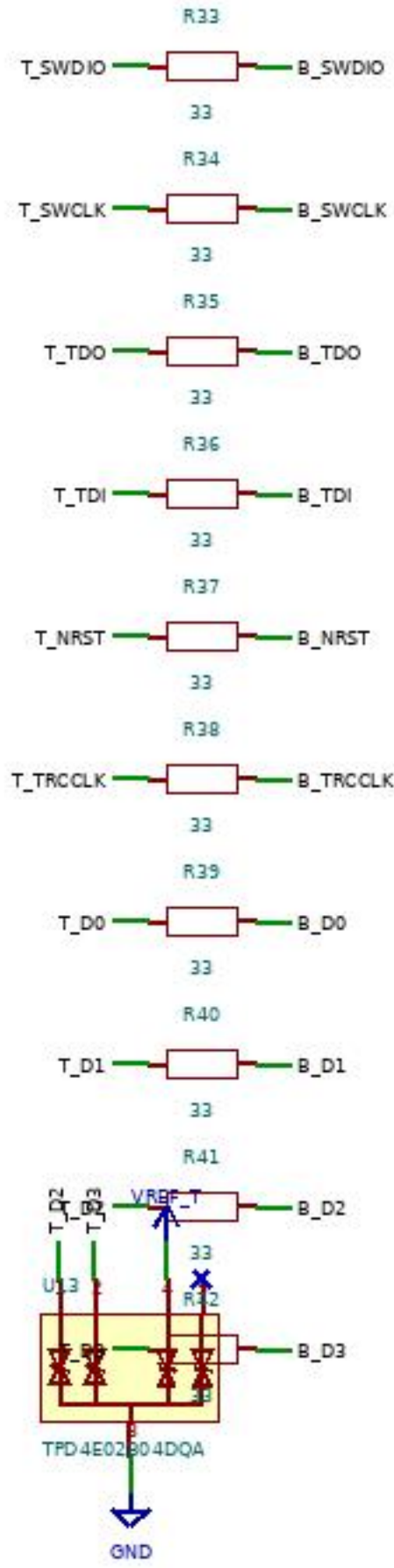
MIPI20 (RISC-V Trace Connectors 1.0 = Cortex Debug+ETM)



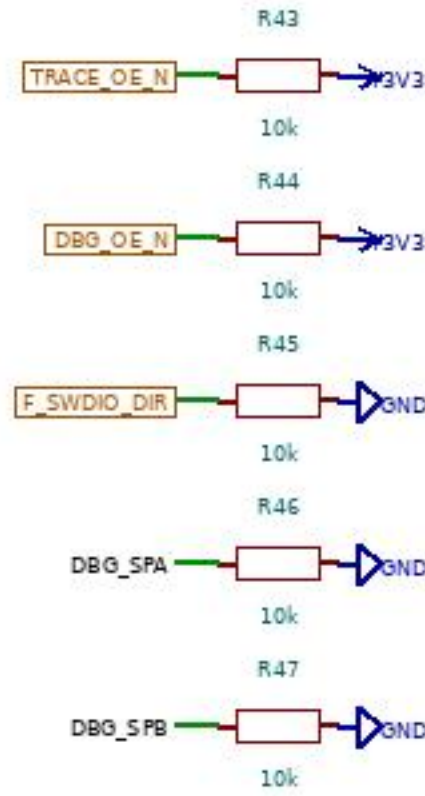
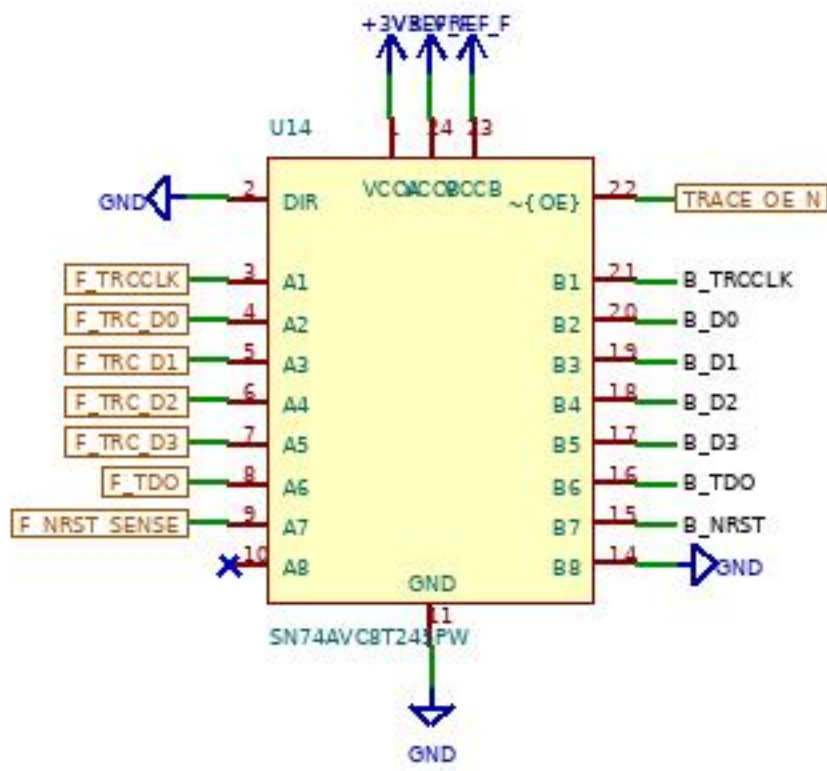
Pin 1 VREF, 2 TMS/SWDIO, 4 TCK/SWCLK, 6 TDO/SWO, 8 TDI, 10 nRESET, 12 TRC_CLK, 14/16/8/20 TRC_DATA[0..3], 9 GNDDetect = GND on probe, 11/13 GND or TgtPwr+Cap (JP1), 7 key/GND.



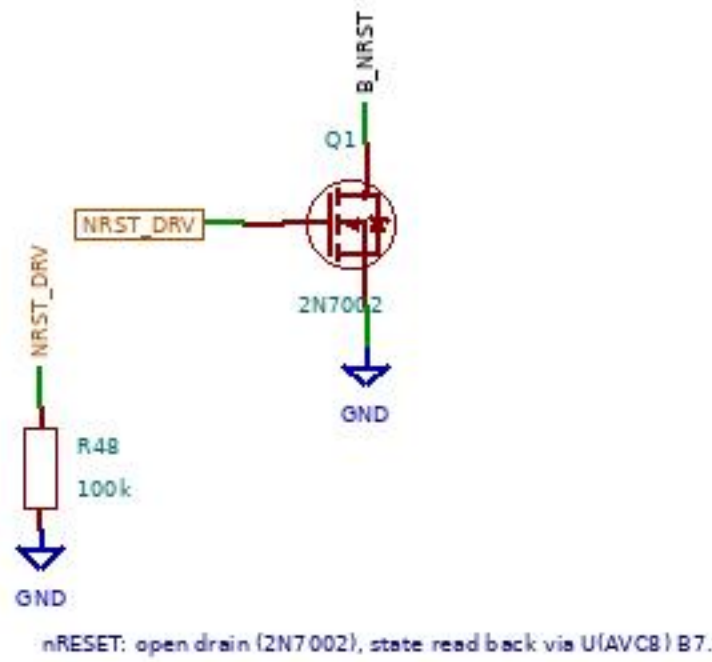
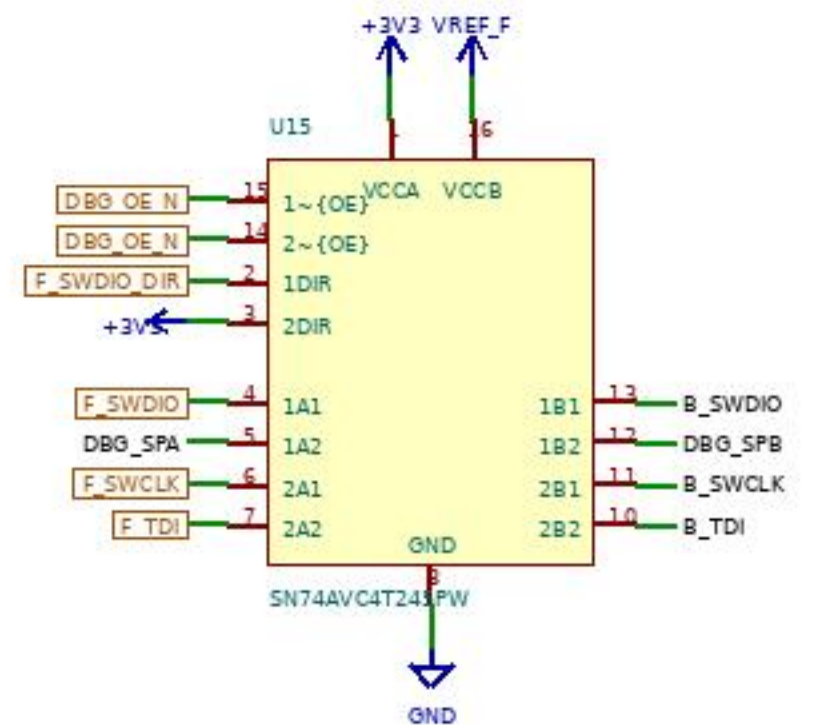
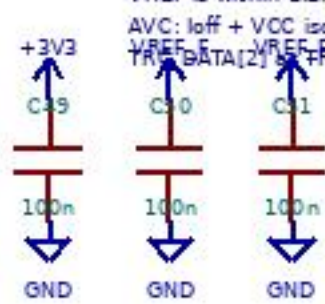
ESD array at the connector, then series R (33 Ohm, tune on prototype).



Level translation (VREF 1.2..3.6 V), nRESET

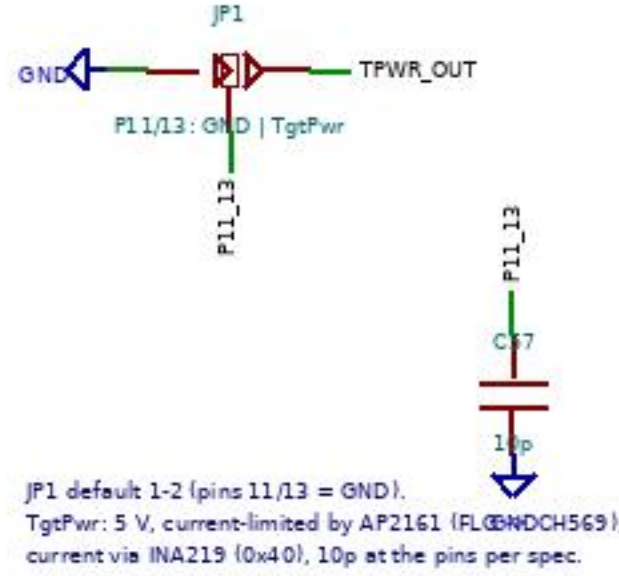
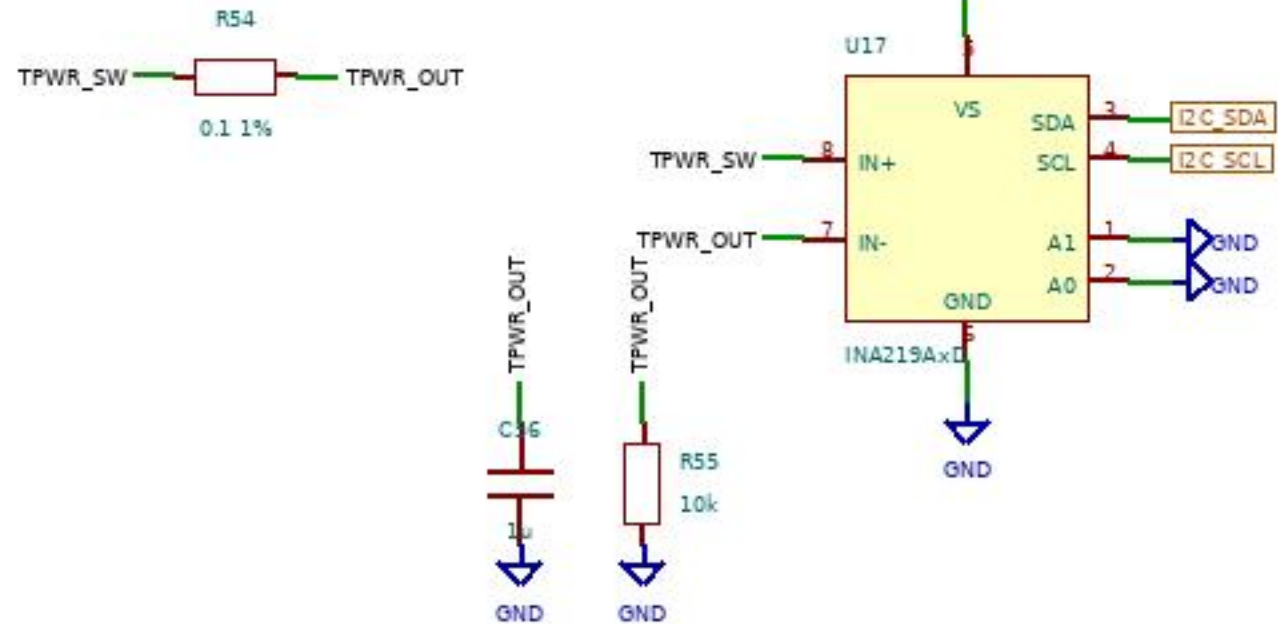
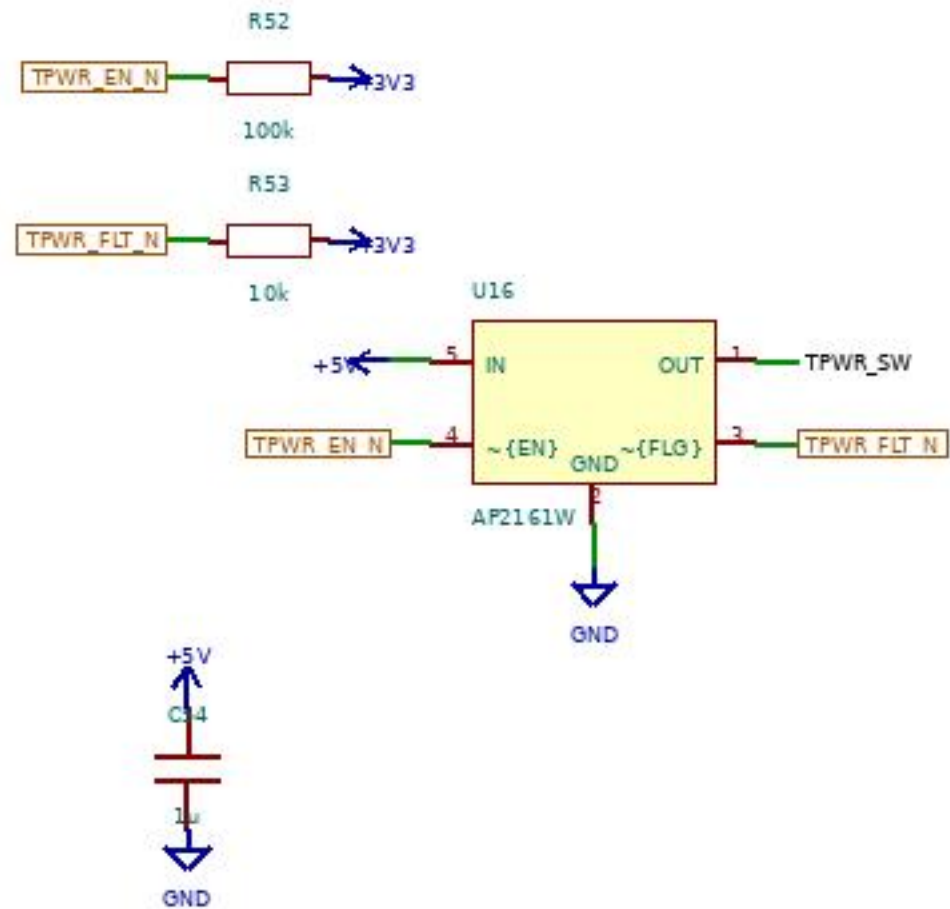


Default after power-up: both translators disabled (OE pulled high), SWDIO direction = target->PFOA. Firmware enables OE only when VREF is within 1.2..3.6 V (ADS1115 AIN0).
AVC: IsoFF + VCC isolation -> Hi-Z when VREF = 0.
VREF_F: VREF_F (IOIN output is not supported in rev 0.1).

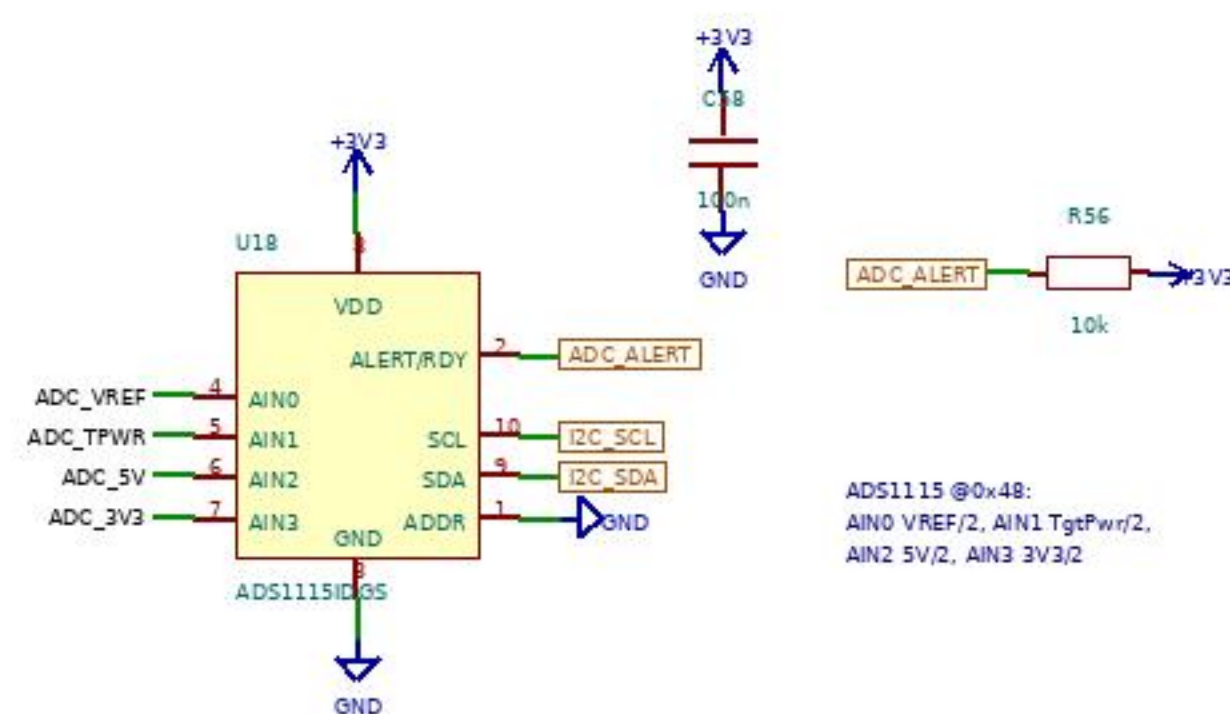
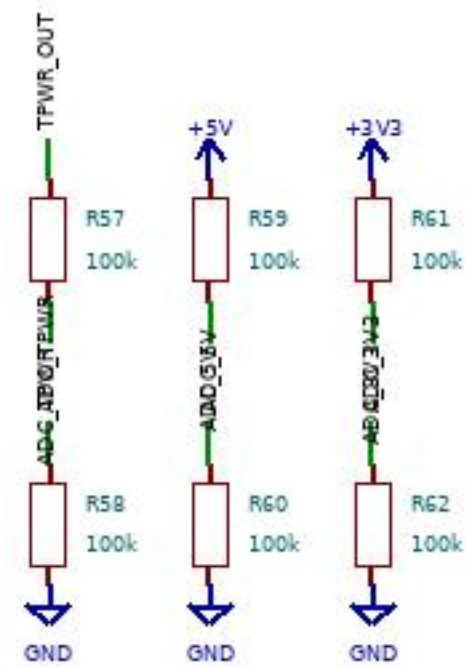


nRESET: open drain (2N7002), state read back via U(AVCB)B7.

Target power (pins 11/13), measurement

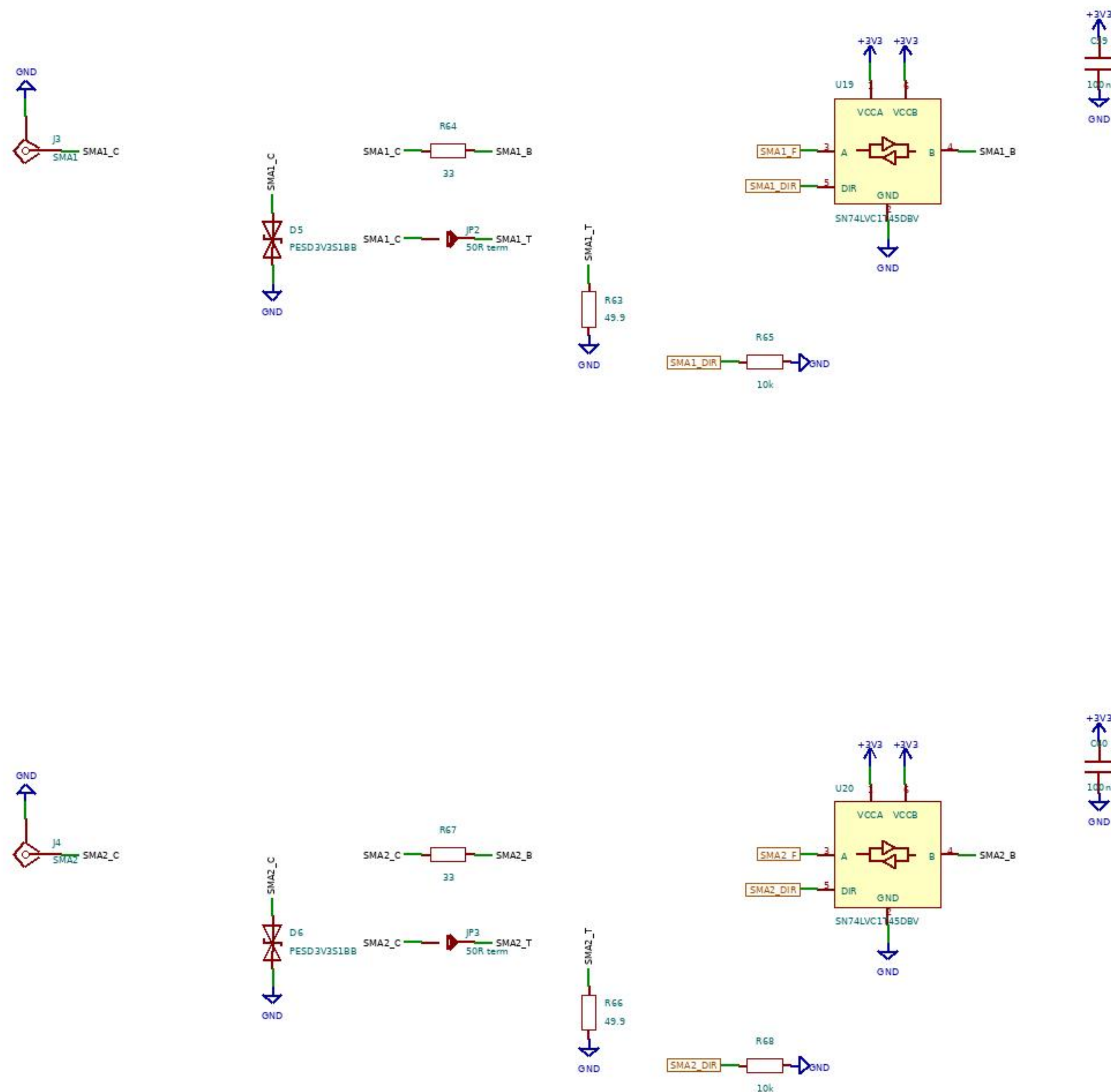


JP1 default 1-2 (pins 11/13 = GND).
TgtPwr: 5 V, current-limited by AP2161 (FLOENCH569), current via INA219 (0x40), 10p at the pins per spec.



ADS1115 @0x48:
AIN0 VREF/2, AIN1 TgtPwr/2,
AIN2 5V/2, AIN3 3V3/2

SMA1 / SMA2: bidirectional 3.3 V LVCMOS (default input)



SMA1 -> FPGA LPLL2_T_in (pin 12): 10 MHz TTL reference from DOROGO or trigger/PFS input; can also drive a trigger out.
SMA2 -> FPGA CLK_T_3 (pin 63): PFS / trigger in or out.
DIR = 0 (default, 10k pull-down): SMA -> FPGA. DIR = 1: FPGA -> SMA.
33 Ohm series + buffer Rout ~ 50 Ohm source termination.
JPx closed: 50 Ohm load for long coax inputs (keep open when driving).
Input level: 3.3 V LVCMOS/TTL only; sine reference is not accepted.